Splash Rules II: Frank–Kasper phases

Every count of the tetrahedrally close-packed metals, exactly, on 18 real phases and 181 layered stackings, and the one number behind them



DICTO · study of 31 October 2026 · results for DICTO, not for publication
DICTO's idea (Splash Rules, 27 October): plot a whole family of exact values and look for one rule that covers the local ones. Second outside run, after the perovskites. Computation with Claude (Anthropic).

Verdict. One number covers the known rules, but the rule is old. Every topological count per atom (tetrahedra, bonds, six-fold bonds, faces) is a straight line in the density of disclination lines, ρ = six-fold bonds per atom, because $\bar{Z} = 12 + 2\rho$ is an identity. DICTO's candidate for a "hexagonality of tetrahedral packing", the share $m = 1 - x_{12}$ of non-icosahedral sites, also works: on every real phase computed (18) and on every further structure listed as known by Deza et al. (13), $\bar{Z} = 66/5 + (2/5)\cdot m$ exactly. This is the Yarmolyuk–Kripyakevich relation of 1974 ($\bar{Z} = 13.6 - 0.4\cdot x_{12}$), and it gives Frank and Kasper's range 13⅓ to 13½ at once. It is not a theorem: three hypothetical structures from a computer search (Dutour Sikirić, Delgado Friedrichs and Deza 2010) leave the line, and they leave it only through one second number, the YK defect a_0: $\bar{Z} = 68/5 - (2/5)x_{12} - (6/5)a_0$. That is the shape of the perovskite Window Law, with a_0 in the place of the Swap defect.
What is new. The Window Law holds in both layered families of Frank–Kasper phases: site counts are lines in the share of tile rows or slabs (window 1), every bond, neighbour, ring and tetrahedron count by kind needs at most neighbouring pairs (window 2), and every pair count fails the Swap Rule with defects in the pattern +1, −2, +1. The c/h stacking of the Laves slabs is invisible to all 119 counts. On the 18 real phases one more exact relation among the six-bond counts holds, unexplained (Section 5). Known before: the identities, the YK line, the two layer constructions and their composition formulas.

1. Set-up

A Frank–Kasper (tetrahedrally close-packed, TCP) structure is cut by its atoms into tetrahedra only. The neighbours of an atom are its Delaunay neighbours, the faces of its Voronoi cell. Each atom has Z = 12, 14, 15 or 16 neighbours; a bond is shared by 5 or 6 tetrahedra; Z = 12 + (number of six-fold bonds). The six-fold bonds form the major skeleton (Frank and Kasper), the disclination network (Nelson): lines through Z14 sites that meet three at a time at Z15 and four at a time at Z16.

Coordinates: AFLOW prototype encyclopedia (CIF files) and the Crystallography Open Database (COD), 18 real phases (Table 1). The topology is read by a periodic Delaunay step on the published coordinates and then checked: every coordination shell is a closed triangulated sphere (Euler number 2), every bond fold is 5 or 6, Z = 12 + #six-fold bonds, no two six-fold neighbours are adjacent in a shell. Every count after that is an integer, and every test below is exact (fractions). Four phases fail the check and are not TCP: χ (α-Mn) and

$Mg_{17}Al_{12}$ (bonds shared by 4 tetrahedra), $Mg_{23}Al_{30}$ and $Mg_{21}Zn_{25}$ (folds 3, 4 or 7). So χ is not Frank–Kasper, as expected.

Two layered families give the splash its spread, built exactly from words:

Square–triangle rows (Sullivan; Sadoc and Mosseri 2016): a word in S (a row of squares) and T (a row of triangles) fixes a square–triangle tiling; its vertices carry Z14 columns, edge midpoints Z12, triangle centres Z15, four Z14 inside each square. S alone is A15, T alone is Z (Zr_4Al_3), ST is H, STT is J. All 62 words checked TCP (all primitive words up to length 7 and 30 random words of 20 rows).

Kagome slabs (Frank and Kasper 1959, read off the AFLOW and COD files here): kagome nets whose hexagons are centred at close-packed letters a_i; each step is $t_i \in \{0, +, -\}$. A step ± is a Laves slab (three-atom sandwich, 6 atoms); a step 0 is a Z slab (two caps and a pair, 7 atoms). + alone is C15, +− is C14, 0 is Z, 0+ is μ, 0+0− is K_7Cs_6. All 119 words checked TCP (all words up to length 6 and 40 random words of 14 slabs). The compositions of I, C, X and Mg_4Zn_7 are those of one Z slab with 2, 3, 5 and 8 Laves slabs.

2. The identities: one number, ρ

> **Identities** (every TCP structure; checked on all 199). Per atom, with ρ = six-fold bonds per atom:
>
> $$\bar{Z} = 12 + 2\rho, \quad \text{bonds} = 6 + \rho, \quad \text{five-fold bonds} = 6, \quad \text{tetrahedra} = 5 + \rho, \quad \text{triangles} = 10 + 2\rho.$$
>
> No tetrahedron has two six-fold edges at one corner; a tetrahedron has 0, 1 or 2 six-fold edges, and two only when they are opposite.

Proof. An atom has 12 five-fold bonds whatever its Z, so five-fold bonds per atom = 12/2 = 6. Its shell has $2Z - 4$ triangles, so tetrahedra per atom = $(2\bar{Z} - 4)/4 = \bar{Z}/2 - 1$. Two six-fold edges ab, ac of one tetrahedron would make b and c adjacent six-fold vertices in the shell of a, which a Frank–Kasper shell does not have. ∎ All of this is classical (Frank and Kasper; Nelson's disclination picture; Coxeter's mean number of tetrahedra per edge, $6 - 12/\bar{Z}$, is the one count here that is not a line).

3. The YK line and its defect

> **Yarmolyuk–Kripyakevich (1974).** In every real Frank–Kasper phase, $6N_{12} = 2N_{14} + 7N_{15} + 12N_{16}$, equivalently
>
> $$\bar{Z} = 68/5 - (2/5)\cdot x_{12} = 66/5 + (2/5)\cdot m, \quad \rho = (3 + m)/5, \quad \text{tetrahedra per atom} = (28 + m)/5.$$
>
> In general $\bar{Z} = 68/5 - (2/5)x_{12} - (6/5)a_0$ with $a_0 = x_{12} - x_{14}/3 - 7x_{15}/6 - 2x_{16}$.

Proof of the general form. Expand both sides with $x_{12} + x_{14} + x_{15} + x_{16} = 1$. ∎ The plane $a_0 = 0$ meets the simplex of shares in a triangle with corners A15 ($\bar{Z} = 27/2$), C15 (40/3) and the Z12:Z15 = 7:6 point (174/13); $\bar{Z}$ is linear, so on the plane $40/3 \le \bar{Z} \le 27/2$, Frank and Kasper's range. The real phases are sums of the three YK units A15 (1:3:0:0), Laves (2:0:0:1) and Z (3:2:2:0) with whole counts, e.g. R = 6 A15 + 24 Laves + 9 Z, Bergman T = 40 Laves +

6 Z (Table 1). In the square–triangle family the units are rows (S = 2 A15 units, T = 1 Z unit), in the kagome family slabs (± = 2 Laves units, 0 = 1 Z unit), so YK is a theorem for both families. For other structures it is not: the computer search found three counterexamples, 7:4:2:2, 6:6:4:0 and 6:8:4:0, with $a_0 = -2/45, -1/24, -2/27$; they sit exactly on $\bar{Z}$ − line = $-(6/5)a_0$ (Figure 1b). The third has $\bar{Z} = 122/9 > 13½$.

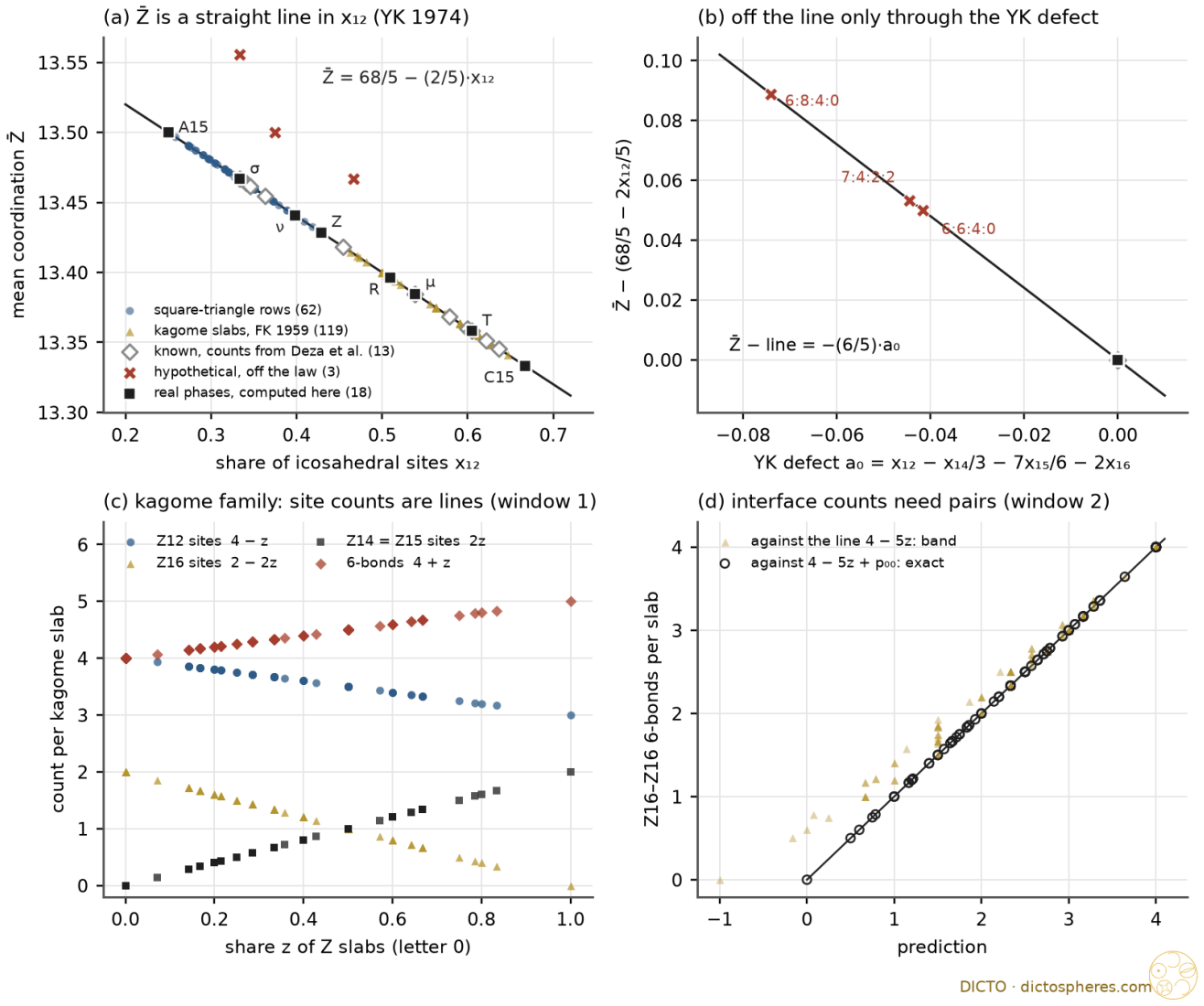


Figure 1. The splash, 199 TCP structures. (a) $\bar{Z}$ against the icosahedral share: every real phase (black squares), every listed known phase (open diamonds) and both families lie on the YK line; three hypothetical structures (red crosses) do not. (b) Their distance from the line is exactly −(6/5) times the YK defect. (c) Kagome family, per slab: site counts and six-fold bonds are lines in the share z of Z slabs. (d) The Z16–Z16 six-fold bonds per slab fill a band against the line 4 − 5z and fall exactly on the diagonal once the share p_{00} of neighbouring Z-slab pairs is added: $4 - 5z + p_{00}$.

4. The Window Law in the layered families

> **Result** (exact on all 62 and 119 stackings). Per tile row (square–triangle) or per kagome slab: every site count and every topological count is a line in the share of S rows or Z slabs (window 1). Every count by kind (bonds by end types, neighbour types, rings of six-fold bonds, tetrahedra by corner types) is fixed by neighbouring pairs of units (window 2). No count needs more. The Laves letter (c or h) changes no count.

Window 1 (per row, per slab): sites S 8 / T 7, slab ± 6 / 0 7; Z12: S 2, T 3, ± 4, 0 3; Z14: S 6, T 2, 0 2; Z15: T 2, 0 2; Z16: ± 2; tetrahedra: S 46, T 40, ± 34, 0 40; six-fold bonds: S 6, T 5, ± 4, 0 5. These are the composition matrices of Sadoc and Mosseri (square–triangle) and of the Laves/Z intergrowth picture (kagome), both known.

Window 2: a count F fixed by a pair of neighbouring units has mean F(TT) + [F(ST) + F(TS) − 2F(TT)]·s + Δ·p, with s the S share, p the share of SS pairs and Δ = F(SS) + F(TT) − F(ST) − F(TS), exactly as in the perovskite study. Kagome: F(LL) + [F(LZ) + F(ZL) − 2F(LL)]·z + Δ·p with z the share of Z slabs and p the share of 00 pairs. Δ is never zero for a window-2 count (else it would be a line). The defects come in the perovskite pattern +1, −2, +1 over the three end-type pairs, so their sum, a window-1 count, is a line:

count	square–triangle (S/T)	Δ	kagome (Laves L / Z)	Δ
six-fold bonds Z14–Z14, Z14–Z15 (Z16), Z15–Z15 (Z16–Z16)	2 + 3s + p, 2s − 2p, 3 − 4s + p	+1, −2, +1	z + p, 2z − 2p, 4 − 5z + p	+1, −2, +1
neighbours of the same pairs (ordered)		+2, −2, −2, +2		+2, −2, −2, +2
tetrahedra 2444, 2445, 2455 (S/T); 2266, 2246, 2244 (L/Z)	36s + 4p, 8s − 8p, …	+4, −8, +4	24 − 30z + 6p, 12z − 12p, 6p	+6, −12, +6
tetrahedra 4444, 4445, 4455 (two opposite six-fold edges)	4s + 2p, …	+2, −4, +2	only 4455: 6z	0

(s, z: shares of S rows and Z slabs; p: share of neighbouring SS or 00 pairs; per row or slab. Tetrahedron kinds by corners, 2 = Z12, 4 = Z14, 5 = Z15, 6 = Z16. Full list: data/results.md.)

Stacking-blindness. 22 Z/L patterns of the kagome family are realised by two or more different Laves stackings; for each, all 119 counts per slab agree. So no count in this census sees whether a Laves slab sits cubic or hexagonal: C15, C14, C36, 9R and the long-period Laves phases have the same census per atom, and so do μ and K_7Cs_6. The skeleton counts agree too; only the embedding of its 4-valent net differs (cubic or hexagonal).

5. The 18 real phases

phase	compound (source)	N	Z12:Z14:Z15:Z16	$\bar{Z}$	tetr./atom	ρ	a_0	YK units A15+L+Z	major skeleton
A15	Cr_3Si (AFLOW)	8	2:6:0:0	27/2	23/4	3/4	0	2+0+0	straight lines, 3 directions
C15, C14, C36	$MgCu_2$, $MgZn_2$, U_2Cr_3Si, $MgNi_2$ (AFLOW)	24, 12, 12, 24	2:0:0:1	40/3	17/3	2/3	0	0+N/3+0	3D net, 4-valent
Laves 9R and long period	$Mg(Li,Zn)_2$, $Mg(Zn,Ag)_2$ (COD 1522206, 2105354, 2105334)	54, 54, 126	2:0:0:1	40/3	17/3	2/3	0	0+N/3+0	3D net, 4-valent
Z	Zr_4Al_3 (AFLOW)	7	3:2:2:0	94/7	40/7	5/7	0	0+0+1	layer net (3-valent) + lines
σ	CrFe type (AFLOW; COD 2106167)	30	10:16:4:0	202/15	86/15	11/15	0	4+0+2	layer net (2,3-valent) + lines
μ	W_6Fe_7 (AFLOW)	39	21:6:6:6	174/13	74/13	9/13	0	0+6+3	3D net (2,4) + layer nets (3)
δ	MoNi (COD 2310246)	56	24:20:8:4	94/7	40/7	5/7	0	4+4+4	3D net (2,3,4)
P	Mo–Ni–Cr (COD 2311049)	56	24:20:8:4	94/7	40/7	5/7	0	4+4+4	3D net (2,3,4) + lines
R	Mo–Co–Cr, $Mn_{85.5}Si_{14.5}$ (COD 2310299, 2107202)	159	81:36:18:24	710/53	302/53	37/53	0	6+24+9	3D net (2,3,4)
ν	$Mn_{81.5}Si_{18.5}$ (COD 2106034)	186	74:80:20:12	1250/93	532/93	67/93	0	20+12+10	3D nets (2,3,4) + lines
T (Bergman)	$Mg_{32}(Al,Zn)_{49}$ (AFLOW)	162	98:12:12:40	1082/81	460/81	55/81	0	0+40+6	3D net (2,3,4)

Per cell; ρ = six-fold bonds per atom; the six-fold share of all bonds is ρ/(6 + ρ). Z, δ and P share $x_{12} = 3/7$ and so share $\bar{Z} = 94/7$, as the YK line demands, though their shares of Z14, Z15, Z16 differ. Not processed: COD 2310313 (P phase; symmetry operations and cell in different settings), COD 1523827 ($Mg_{51}Zn_{20}$; degenerate points).

One more exact relation. Let σ be the number of six-fold bonds inside an atom's coordination shell. On all 18 real phases and all 181 family members,

$$\Sigma_{\text{Z14 sites}}\,(\sigma - 4) + 2\cdot\Sigma_{\text{Z12 sites}}\,(6 - \sigma) = 8\cdot N_{15}.$$

A15: σ = 4 at Z14 and 6 at Z12, no Z15. Bergman T: 12·2 + 2·36 = 96 = 8·12. The identities fix only the sum of all σ (each six-fold bond lies in six shells, so Σσ = 6ρN), so this is a second, independent relation. It is not explained here. Because every structure with coordinates lies on the YK plane, the relation is fixed only up to adding a multiple of YK, and it cannot be tested on the three counterexamples (no coordinates published in reach). Three more exact relations tie the neighbour-type counts to the shell counts on the same set (data/relations2.txt): the local census of the real phases spans only 8 dimensions where identities and YK leave 12. Likely reason: all of them are built from a few motifs. Status: observed, open.

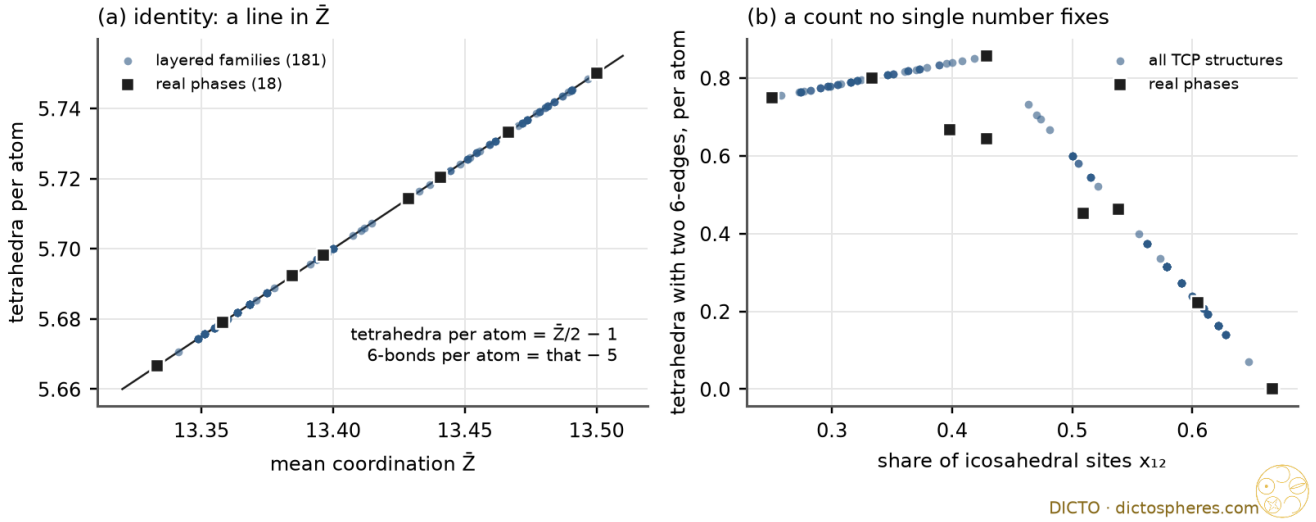


Figure 2. (a) Tetrahedra per atom against $\check{Z}$, all 199 structures: the identity $\check{Z}/2 - 1$. (b) Tetrahedra with two (opposite) six-fold edges against x_{12}: two lines for the two families, the real phases off both. No single number fixes this count.

6. Prior art: sources checked

Reference library (ask-library, six searches: Frank–Kasper polyhedra Z14 Z15 Z16; Yarmolyuk–Kripyakevich mean coordination; disclination network major skeleton Nelson; space fullerenes Frank–Kasper enumeration; Shoemaker kagome layers σ μ Laves stacking; Sadoc–Mosseri square–triangle decoration): no Frank–Kasper paper in the library; hits were tetrahedron packings (Kallus–Elser, Haji-Akbari et al. 2009), Mosseri–Sadoc tile papers (Papadopolos–Ogievetsky 1999, Koca et al. 2020), stacking-disorder papers (Ganesh–Nasr 2023, Hart–Hansen–Kuhs 2021). Sadoc and Mosseri 2016 is in the catalogue only and was read from arXiv. **Read**: Sadoc and Mosseri, Quasiperiodic Frank–Kasper phases derived from the square-triangle dodecagonal tiling, arXiv 1609.01624 (pp. 1–7: major skeleton, the Z16-free decoration, composition matrix, $\bar{z} = (108y + 47)/(8y + 7/2)$, Z16 stripe decoration); Deza, Delgado Friedrichs and Dutour Sikirić, talk "Space fullerenes" (table of 24 known structures with Z12:Z14:Z15:Z16, the YK conjecture $a_0 = 0$, counterexamples 7:4:2:2, 6:6:4:0, 6:8:4:0, Nelson–Spaepen bound 13½ broken, Rivier–Aste conjecture, Laves stacking symbols ±1 and Frank–Kasper's 0, ±1); Gvozdetskyi, Gladyshevskii and Hlukhyy, Chem. Met. Alloys 7, 165 (2014) (YK equations $\langle CN \rangle = 13.6 - 0.4 \cdot C_{12}$ and $C_{14} - C_{16} = 1.4 - 2.6 \cdot C_{12}$, structural formula $(R_3X)_i(PX_2)_j(Q_2R_2X_3)_k$, Hellner–Pearson condition, counterexamples); Dutour Sikirić, Delgado Friedrichs and Deza, Acta Cryst. A 66, 602 (2010) (abstract only; PDF embargoed). AFLOW and COD entries as in Table 1. **Web searches**: "Dutour Sikirić Delgado-Friedrichs Deza space fullerenes … Yarmolyuk Kripyakevich"; "Frank-Kasper structures Yarmolyuk Kripyakevich fractions Z12 Z14 Z15 Z16 A15 C15 Zr4Al3 linear combination"; "Dutour Sikirić Frank-Kasper structures 84 data files" (no data files found). **In-house**: Splash Rules (27 Oct), Splash Rules II perovskites (31 Oct, Window Law), #57, #66 (Swap ⇔ line). **Classical, cited not read**: Frank and Kasper 1958, 1959; Yarmolyuk and Kripyakevich 1974; Nelson 1983 and Nelson and Spaepen 1989; Shoemaker and Shoemaker 1986; Sinha 1972; Sullivan (square–triangle construction); Hellner and Pearson; Rivier and Aste 1996; Coxeter (mean edge number).

7. Conclusion and status

Conclusion: in Frank–Kasper phases one number, the disclination density, fixes every topological count by identity; on all real phases the non-icosahedral share fixes it too (the YK line, 1974), and the only way off that line is the YK defect; inside the layered families the perovskite Window Law holds with windows 1 and 2 and the Swap pattern +1, −2, +1. **Status:** done for counts (study 2026-10-31-splash2-frank-kasper). Open: the shell relation of Section 5, which needs a structure off the YK plane to test (the counterexamples of the 2010 search, if their coordinates can be obtained).